

Protein-Protein Interaction in both Human and E-Coli Bacteria

Introduction:

Studying protein protein interaction is very important because proteins in a cell work together to do jobs. If a gene has to be activated or even if the gene has to replicate or whatever it may be, proteins work together by signaling each other. Diseases occur in our body due to wrong functioning of proteins so it is very important for us to study about proteins. In this project we will be collecting the dataset of human as well as e-coli bacteria so we can understand about diseases , design drugs and know how a cell works.

We are using Machine Learning and Deep Learning to predict if two proteins interact or not. We will be using ML models like Random Forest, Support Vector Machines, and Gradient Boosting. These models work well when we give the features of the proteins whereas DL models learn the patterns directly from the protein sequences.

Objective:

- We collect protein sequences from humans and e-coli and clean them by removing invalid or repeated sequences.
- Make dataset of protein pairs for predicting interactions
- Use ML and DL models to predict if proteins interact

Novelty:

This project is novel because we constructed two separate high-confidence protein-protein interaction datasets for both Human and E. coli. We applied strict filtering using a STRING confidence score ≥ 900 to ensure biological reliability. We also performed sequence cleaning, duplicate removal, and class balancing to prepare a robust machine learning-ready dataset. The comparative cross-species dataset construction adds research value.

Retrieving of Data:

We have collected the datasets from the UniProt and String database.

- **UniProt:**

We have collected fasta files which contain the protein sequences and we have collected protein sequences for human and e-coli bacteria. Each sequence has a header and the amino acid sequence. Header data include UniProt ID, protein names, and species. Since this is the raw data we will remove invalid amino acids, duplicates and unrealistic lengths. After cleaning the dataset we will divide the dataset into interacting and non interacting pairs based on the score of interaction. The dataset we collected from UniProt are reviewed so there is sanity that the sequences are reliable and high quality for analysis.

- **String:**

We have collected three types of files from the String database: aliases, links, info. So the aliases file is very much needed because the same protein may have different ids in different databases so to map those ids we need this aliases file. Aliases file usually has the mapping information .

The next file is a link file so this file contains the protein pair which are interacting along with the confidence score. So this links file is the source for interacting pairs and for non-interacting pairs we divide this links file in such a way that we take a threshold and Negative pairs were generated by randomly pairing proteins that are not present in the high-confidence interaction list. These pairs are assumed to be non-interacting and labeled as 0. Finally the info file which contains the metadata about each protein in the as protein name, description , sequence length, and functional information.

Data Cleaning:

These codes focus on preparing clean and reliable datasets for protein-protein interaction analysis in both *E. coli* and human. The first two codes read the raw protein sequences from UniProt FASTA files and extract important information like UniProt ID, sequence, header, and sequence length. They then remove any sequences containing invalid amino acids, remove duplicates, and filter out sequences that are too short or too long, keeping only proteins between 50 and

1500 amino acids. After cleaning, the sequences are saved in both FASTA and CSV formats, providing a structured and high-quality dataset ready for further analysis.

The third code deals with interaction data from the STRING database for human proteins. It reads a file containing protein pairs along with a combined confidence score, and filters the dataset to keep only **high-confidence interactions** (score $\geq$ 900). This ensures that only reliable positive protein pairs are retained, which is important for generating accurate labels for training predictive models. The cleaned interaction data is then saved as a CSV file to be used alongside the cleaned protein sequences.

Redundant interaction pairs were removed by converting protein pairs into order-independent format (A-B and B-A treated as same). This reduced 62,588 positive samples to 31,294 biologically unique interactions for E. coli. This step ensures that duplicate interaction information does not bias the machine learning model. Sequence duplicates were also removed based on UniProt ID uniqueness.

We built a Positive Protein Interaction (PPI) Dataset for E-coli . In simple terms it prepares data to train a machine learning model, to predict whether two proteins interact or not . First it loads high - confidence protein interaction pairs from the STRING database . These pairs represent proteins that are known or predicted to interact with each other . However, STRING uses its own protein ID system , which cannot directly be used with sequence datasets .

The code loads a STRING alias file to convert STRING protein IDs into UniProt IDs. This is needed because STRING uses its own ID system, but protein sequences are stored using UniProt IDs. Since UniProt IDs are standard and widely used, they help us correctly find the protein sequences. The code keeps only the valid UniProt mappings and removes the unnecessary ones.

Next, it loads a cleaned protein sequence dataset that contains UniProt IDs and their corresponding amino acid sequences. It then ensures that only proteins with available sequences are kept. This step removes incomplete or invalid data.

After that, the code creates a mapping dictionary to convert STRING protein IDs into UniProt IDs. Using this mapping, it replaces STRING IDs in the interaction dataset with UniProt IDs . Any interaction pair that cannot be mapped properly is removed to maintain data quality .

Once the correct UniProt IDs are assigned, the code attaches the actual amino acid sequences of both interacting proteins . Again, it removes any rows where sequences are missing . The same process is done for the human data set also .

This code generates a negative protein–protein interaction dataset for E-coli . It first loads the previously created positive interaction dataset and extracts all unique proteins. Then, it randomly selects pairs of proteins that are not present in the known positive interaction list and treats them as non-interacting pairs. These pairs are labeled as 0 (negative class). The code then attaches the corresponding protein sequences and saves the final negative dataset as a CSV file.

In simple terms, this script creates artificial non-interacting protein pairs so that a machine learning model can learn to distinguish between interacting (label = 1) and non-interacting (label = 0) proteins.

Dataset Statistics:

For E. coli:

Total unique proteins after cleaning: 3877
Total positive interaction pairs: 62588
Total unique biological interactions: 31294
Final balanced dataset size: 125176

For Human:

Total positive interaction pairs: 172230
Final balanced dataset size: 344460

Negative Protein Interaction Dataset Generation (Human):

This code generates a negative protein-protein interaction dataset for human proteins. It first generates all the positive interaction dataset and then randomly pairs proteins that are not known to interact and treats them as non-interacting pairs. The number of negative pairs generated is made the same as the number of positive pairs to maintain class balance. It also adds the amino acid sequences for each protein and labels these pairs as 0 (non-interacting). Then the final negative interaction dataset is saved as a CSV file.

Balanced Dataset Construction for Human Proteins:

This code generates the final dataset for E.coli protein interactions that can be used for machine learning. It loads the positive and negative protein pairs, keeps only the important information like protein IDs, sequences, and labels, and then combines them into one dataset. The dataset is shuffled randomly to mix the data, and the final balanced dataset has equal numbers of interacting (label = 1) and non-interacting (label = 0) protein pairs.

Balanced Dataset Construction for Human Proteins:

This code generates the final balanced dataset for protein-protein interaction prediction. It takes positive and negative pairs and then combines them into one dataset, and shuffles the rows to mix the data. The final dataset is saved as a CSV file and can be used to train machine learning or deep learning models.

Dataset Specificity and Validation

The dataset is specific to true protein-protein interactions because only high-confidence STRING interactions (score ≥ 900) were selected. Such interactions are supported by experimental evidence, curated databases, co-expression analysis, and computational predictions. Additionally, only reviewed UniProt protein sequences were used, ensuring biological reliability. Sequence length filtering removed fragments and abnormal proteins, improving dataset quality

Conclusion:

In this project, we collected protein data from humans and *E. coli*, cleaned it, and created datasets of interacting and non-interacting protein pairs. We combined and balanced the data to make it ready for machine learning and deep learning models. This prepared dataset can now be used to predict protein-protein interactions which helps in understanding cellular processes and disease mechanisms.

